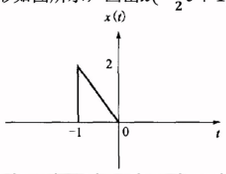
Title: 信号的自变量变换

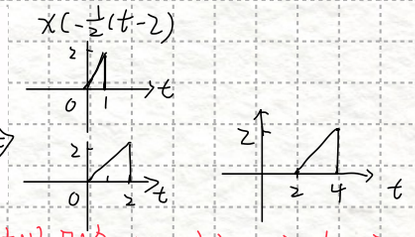
Date: / /

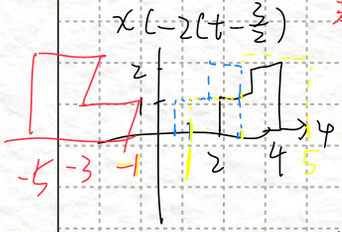
① 化成标准形式
② 首先有负号都翻转
③ |系数|>1 压缩，|系数|<1 拉伸
④ 左加右减

已知$x(t)$的波形如图所示，画出$x(-\frac{1}{2}t+1)$的波形



$\Rightarrow$





若已知 $x(-\frac{1}{2}t+1)$，则令 $u=-\frac{1}{2}t+1 \Rightarrow t=?$ 反推 $x(t)$

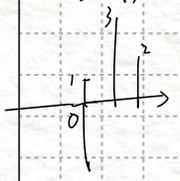
一阶后向差分 $\nabla f[k]=f[k]-f[k-1]$

N阶……$\nabla^n f[k]=\nabla\{\nabla^{n-1}f[k]\}$

一阶前向差分 $\Delta f[k]=f[k+1]-f[k]$

$$x[n]=\sum_{k=-\infty}^{+\infty}x[k]\delta[n-k]$$

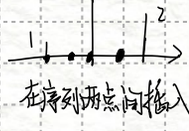
尺度变换



$\Rightarrow$

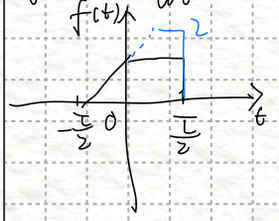
$$f[k/m]=\begin{cases}f[k/m] & k\text{是}m\text{整数倍}\\ 0 & \text{其它}\end{cases}$$

内插

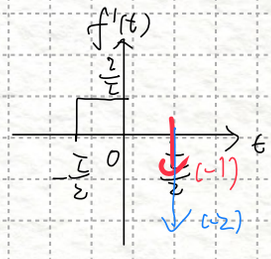


在序列两点间插入$M-1$个点

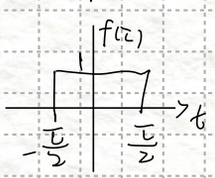
微分：$f'(t)=\frac{df(t)}{dt}$



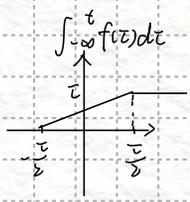
$\Rightarrow$



积分：$\int_{-\infty}^{t}f(\tau)d\tau$



$\Rightarrow$



@自由笔记